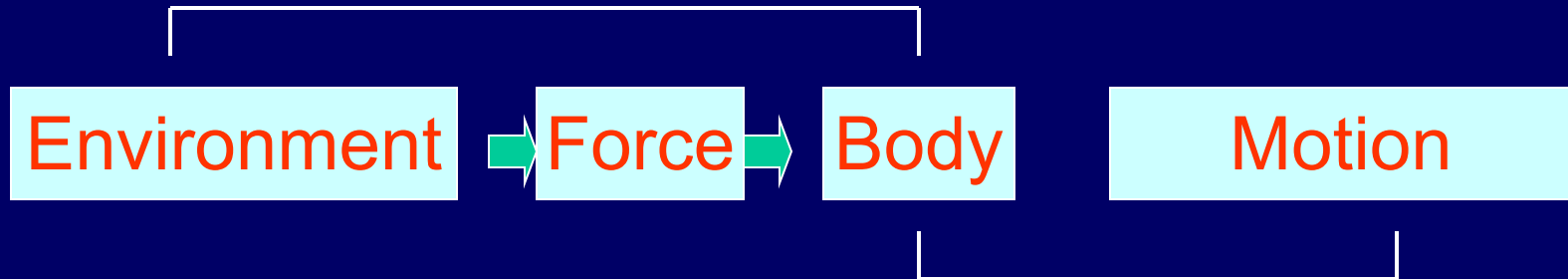


Chap. 3

Force and Newton's law

Force and Motion

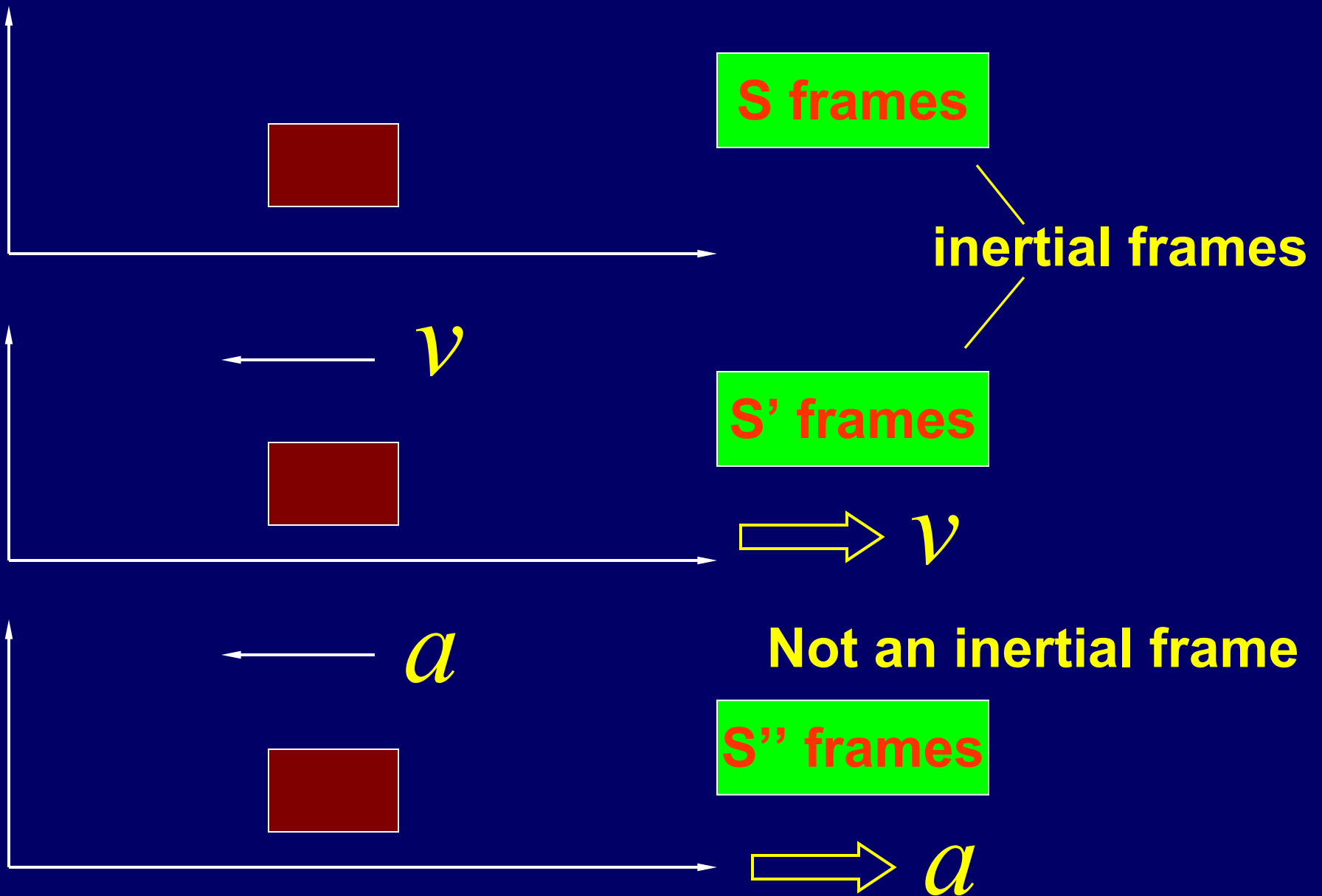
The force laws



The motion laws

We consider a body on which no net force acts. In a reference frame, if the body is at rest, it will remain at rest; if the body is moving with constant velocity, it will continue to do so.

Such reference frame in which Newton's first law is valid is called the **inertial frame**.



Same force acting on

Mass

$$F = ma$$

$$\frac{m_x}{m_{std}} = \frac{a_{std}}{a_x}$$

Newton's Second Laws

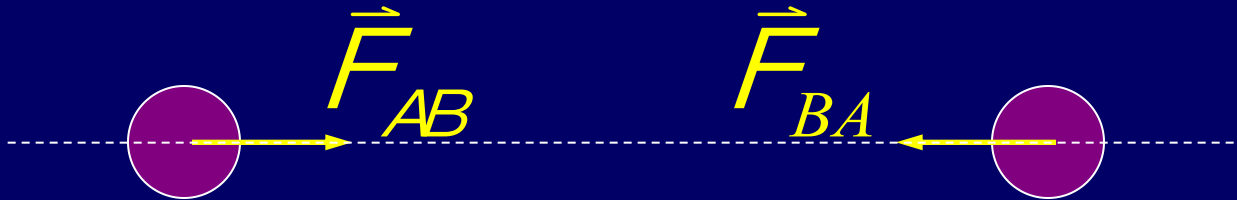
$$\Sigma \vec{F} = m\vec{a}$$

$$\Sigma F_x = ma_x$$

$$\Sigma F_y = ma_y$$

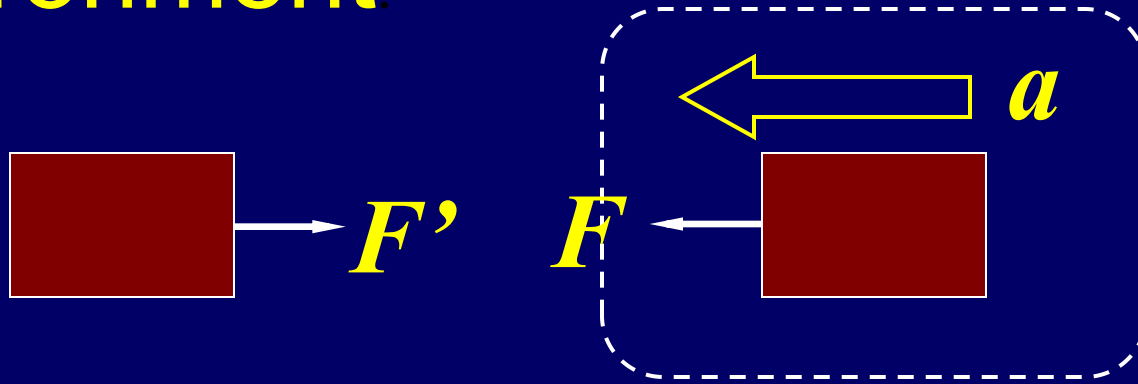
$$\Sigma F_z = ma_z$$

Newton's Third Laws



$$\vec{F}_{AB} = -\vec{F}_{BA}$$

environment.



The force is an interaction between
a body and its environment.

→ If $F=0$, the motion of the body will
remain unchanged.

1st Newton's law

The force acts on a body that will accelerate or decelerate.

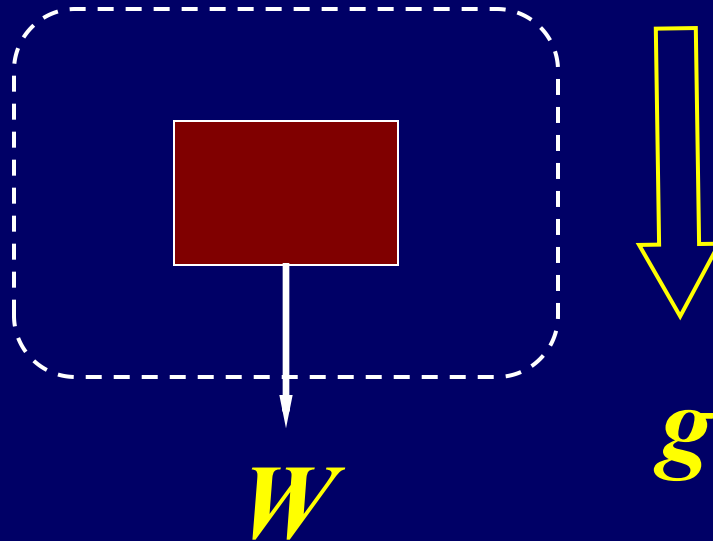
$$\vec{F} = m\vec{a}$$

2nd Newton's law

$$\vec{F} = -\vec{F}'$$

3rd Newton's law

$$W=mg$$



environment

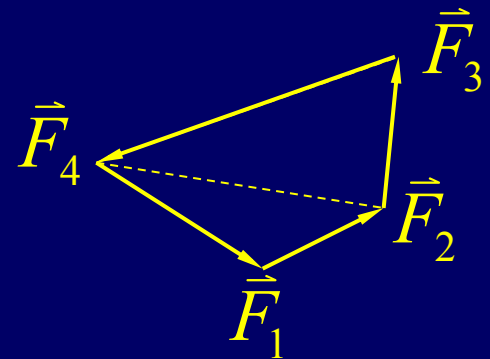
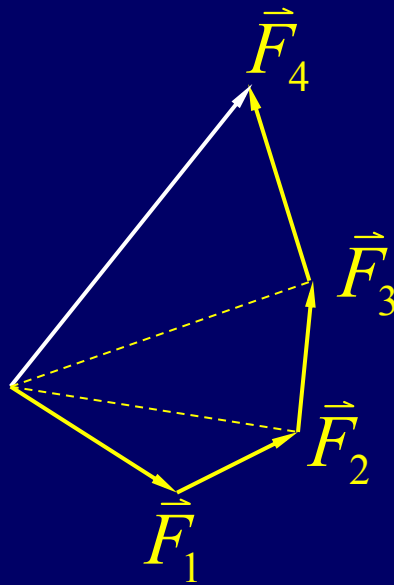
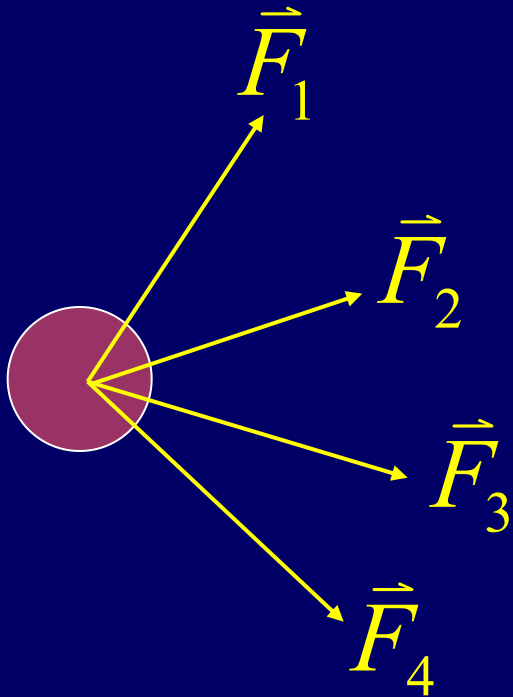


The force is a vector $\sum \vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4$

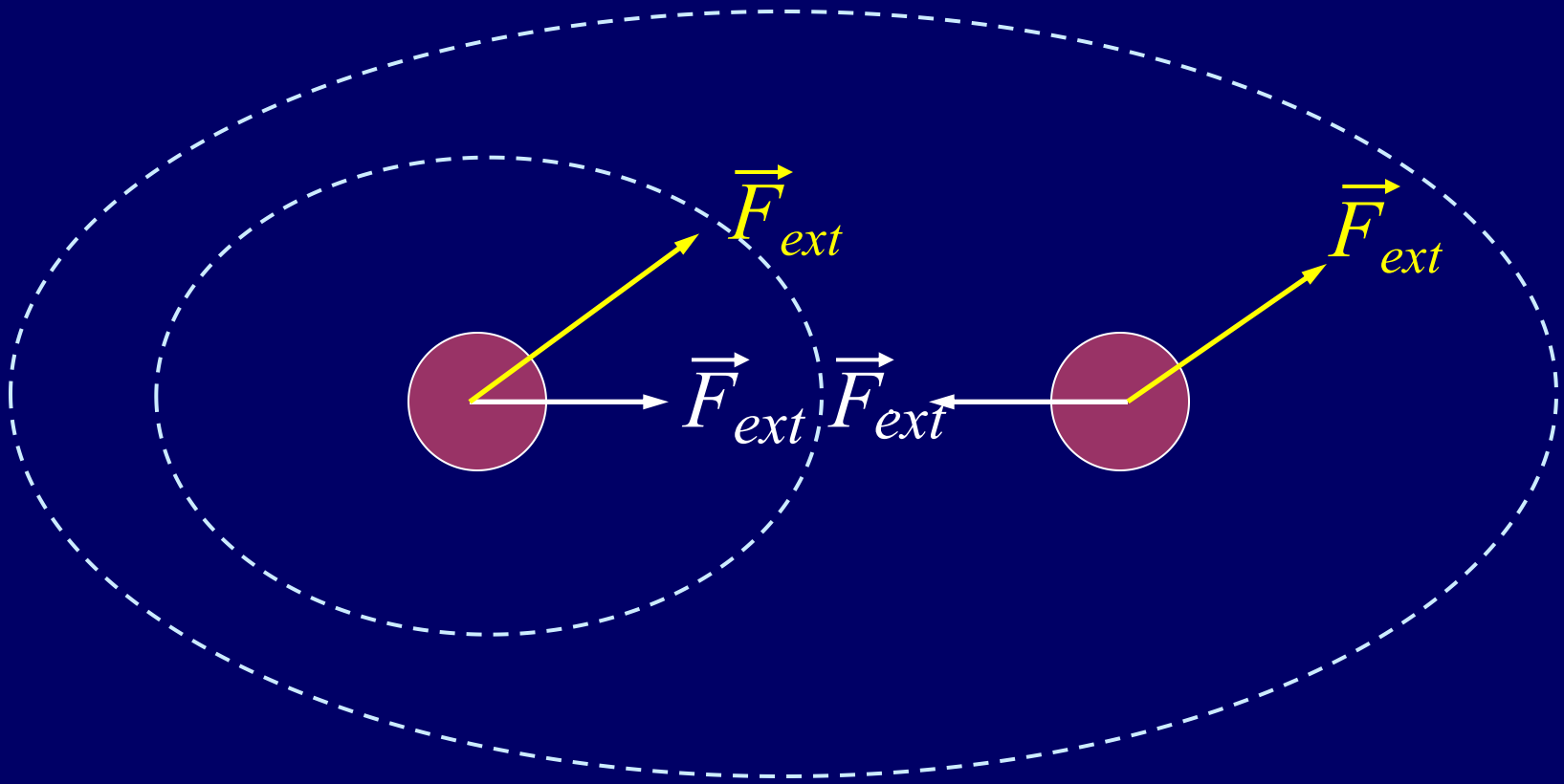
$$\vec{F}_1 \neq 0, \vec{F}_2 \neq 0, \vec{F}_3 \neq 0, \vec{F}_4 \neq 0, \quad \sum \vec{F} = 0$$

No net force

$$\vec{a} = 0$$



External force and internal force



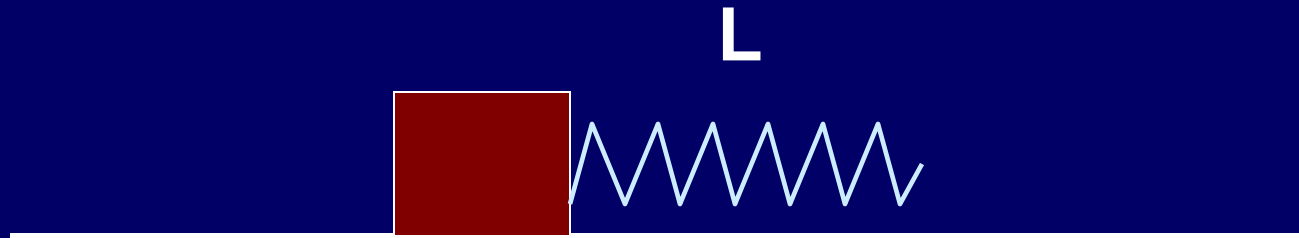
UNIT OF FORCE

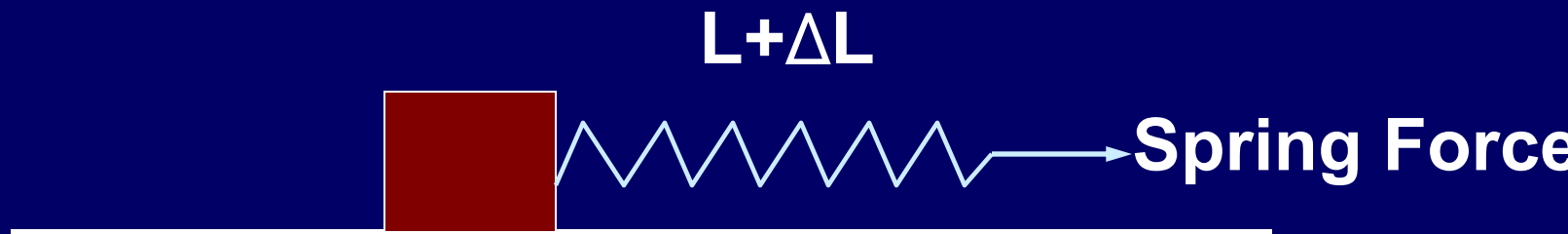
newton, N

$$1\text{N} = 1\text{ Kg}\cdot\text{m}/\text{s}^2$$

$$[F] = \text{ML}/\text{T}^2$$

Spring Force





Tension force T

The ideal string :

very thin

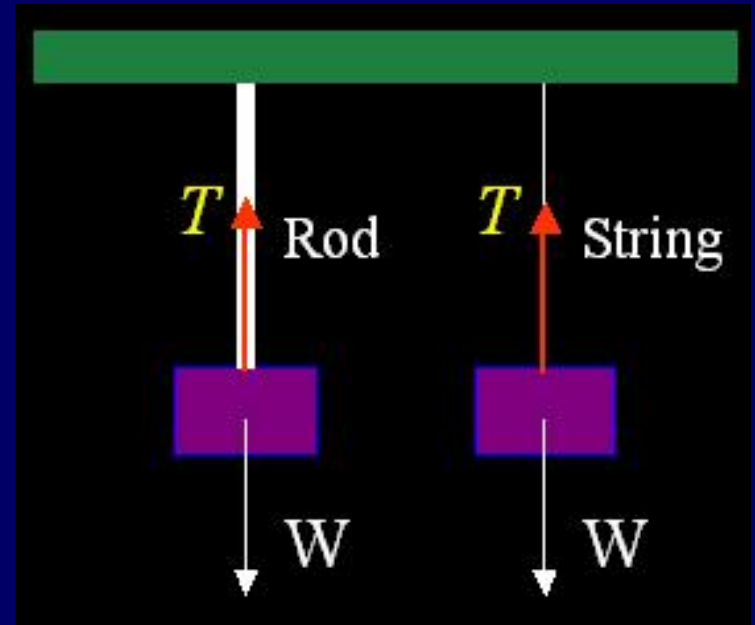
negligible mass



For a massless and non-stretch string (or rod), there is always an action-reaction pair at any point in the string (or rod).

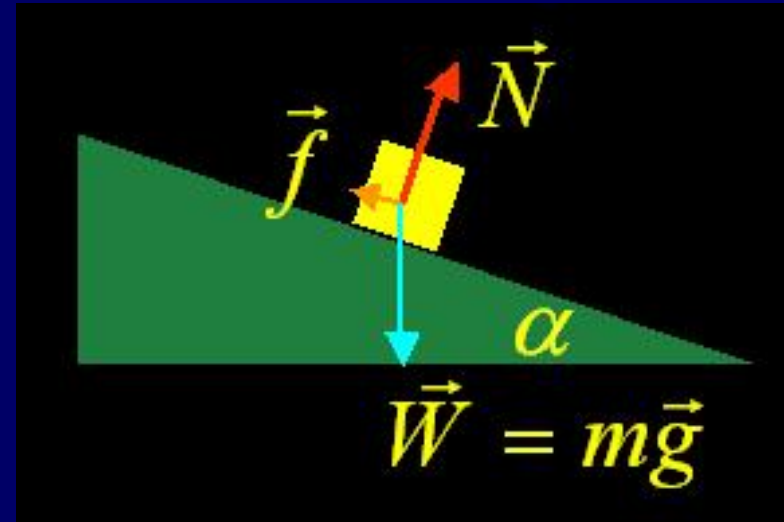
$$\vec{T} = -\vec{W}$$

$$\vec{W} = m\vec{g}$$



▣ Normal force N

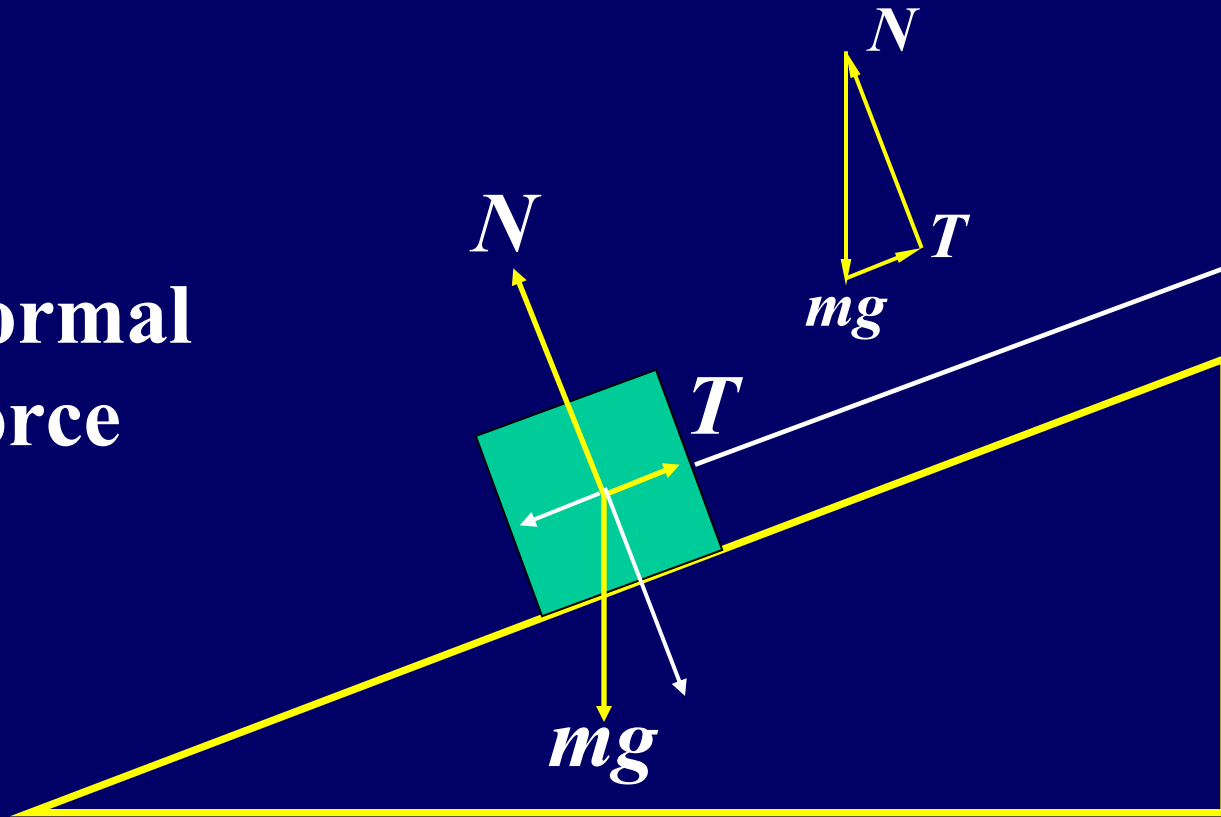
$$N = mg \cos \alpha$$



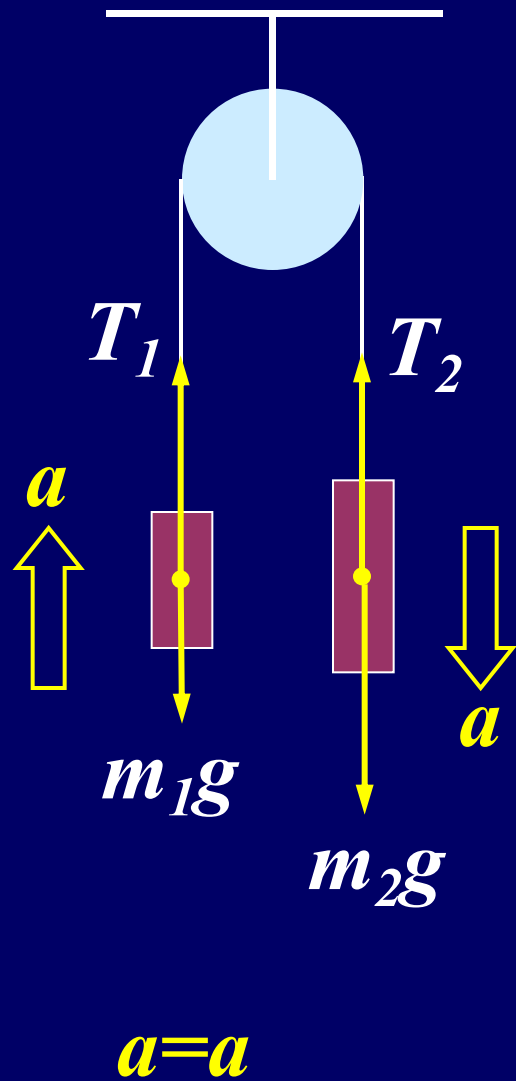
The normal force is always perpendicular to the surface.

T and N are all elastic forces. They are passive forces.

**Normal
Force**



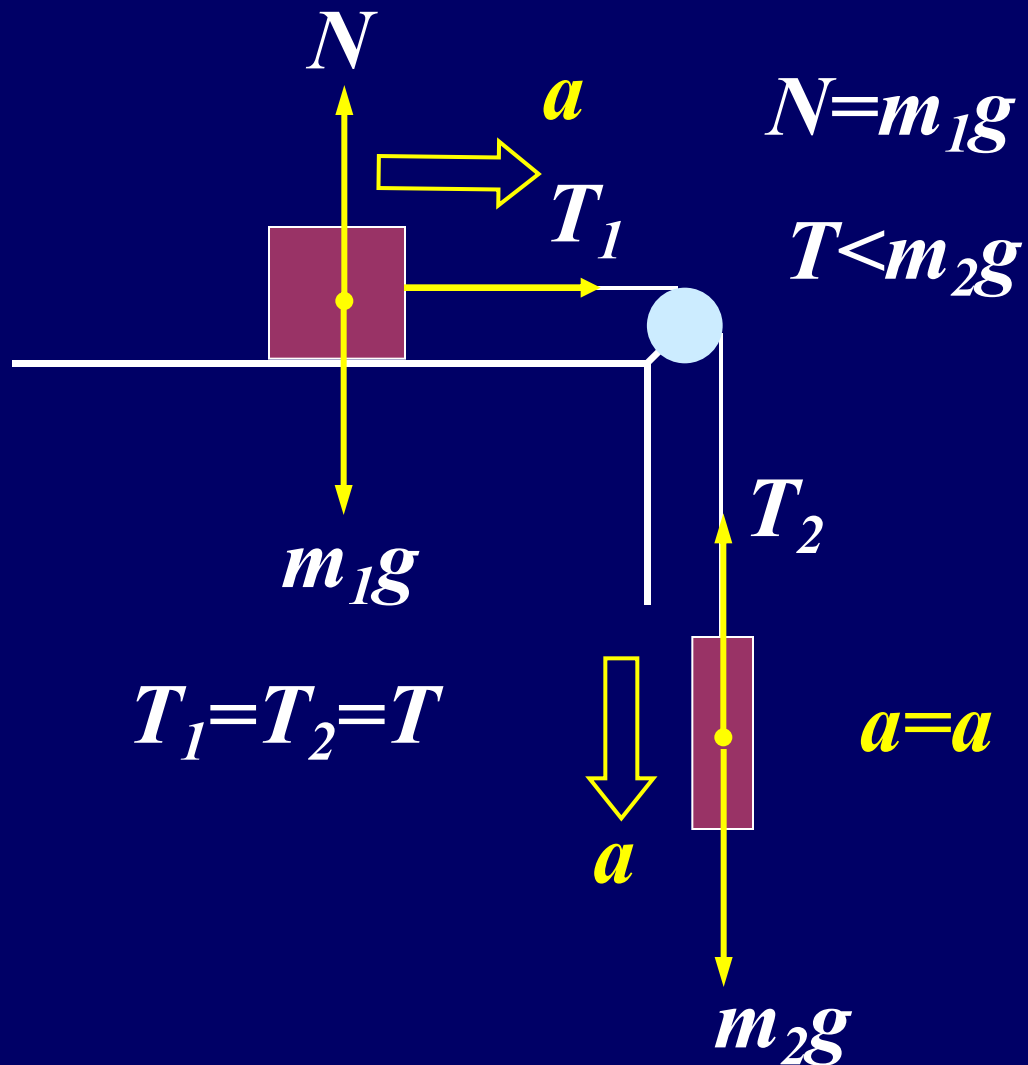
$$\Sigma \vec{F} = 0 \quad \vec{a} = 0$$



$$T_1 = T_2 = T$$

$$m_2g > T$$

$$m_1g < T$$



$$N = m_1g$$

$$T < m_2g$$

$$T_1 = T_2 = T$$

Gravitational Force



$$F = \frac{Gm_1m_2}{r^2} \quad G \text{ gravitational constant}$$

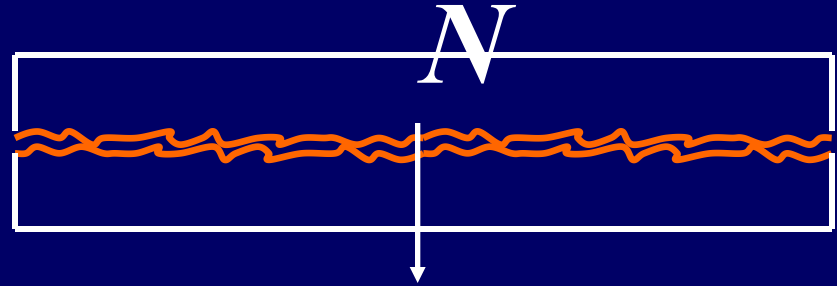
$$\vec{F} = F\hat{u}_r \quad \vec{F} = -\frac{Gm_1m_2}{r^2}\hat{u}_r \quad \vec{F} = -\frac{Gm_1m_2}{r^3}\vec{r}$$

$$F_s = \frac{GMm}{R^2} = mg$$

frictional force

Kinetic friction

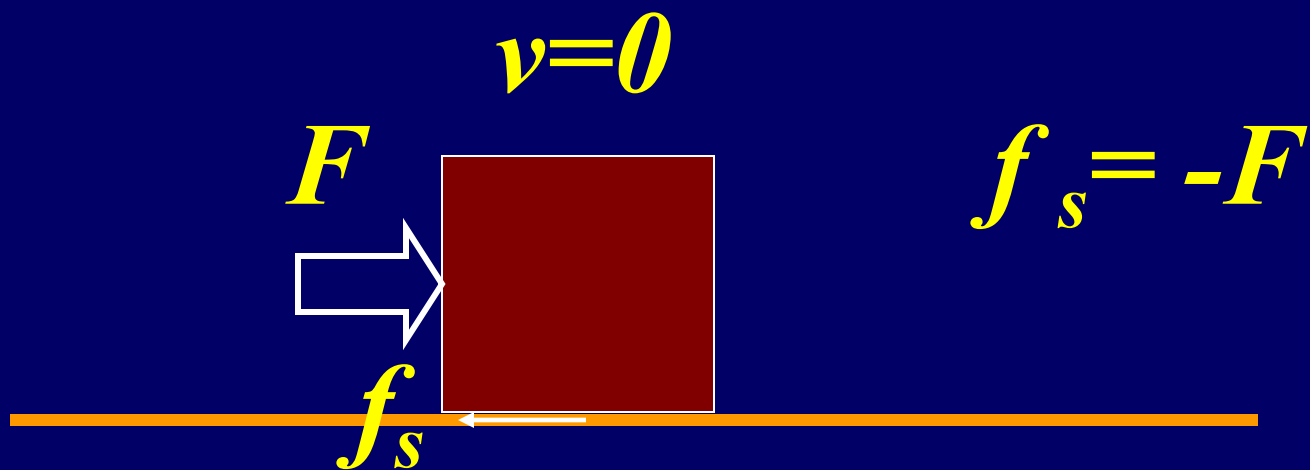
Static friction



$$f_k = \mu_k N$$

$$f_s = - \Sigma F_{ext} \leq \mu_s N$$

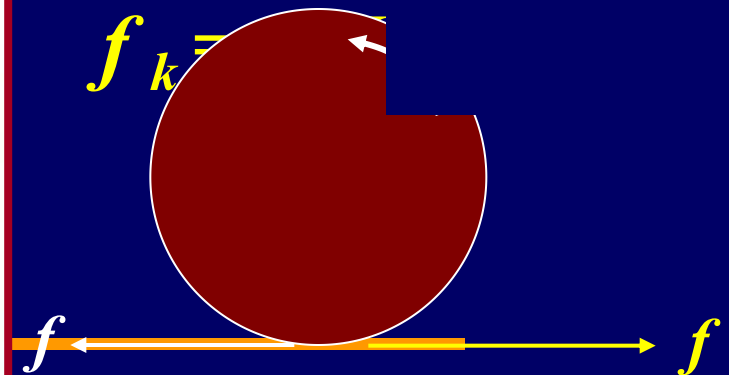
$$\mu_k < \mu_s$$



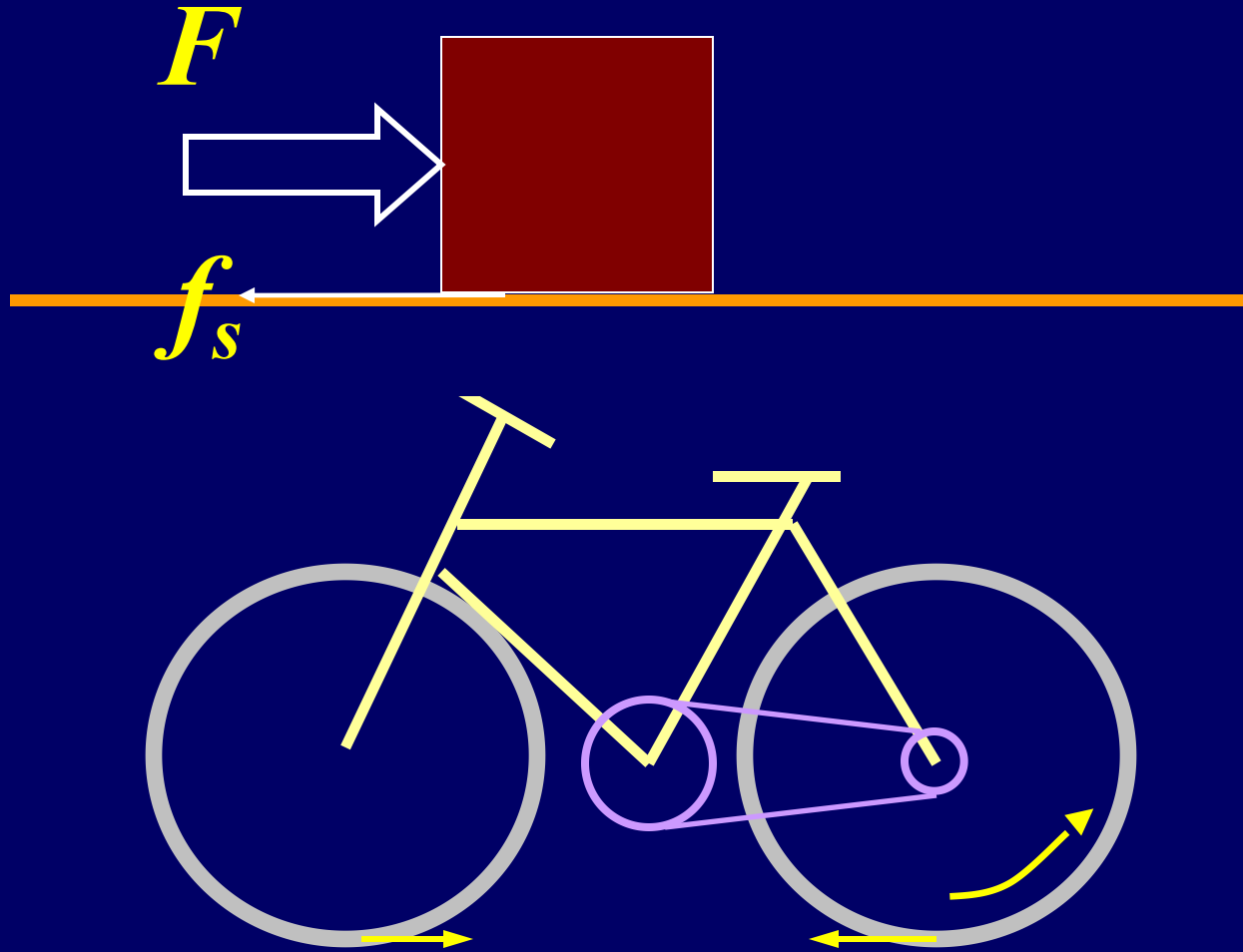


**With ABS, the wheel
rotates without slipping**

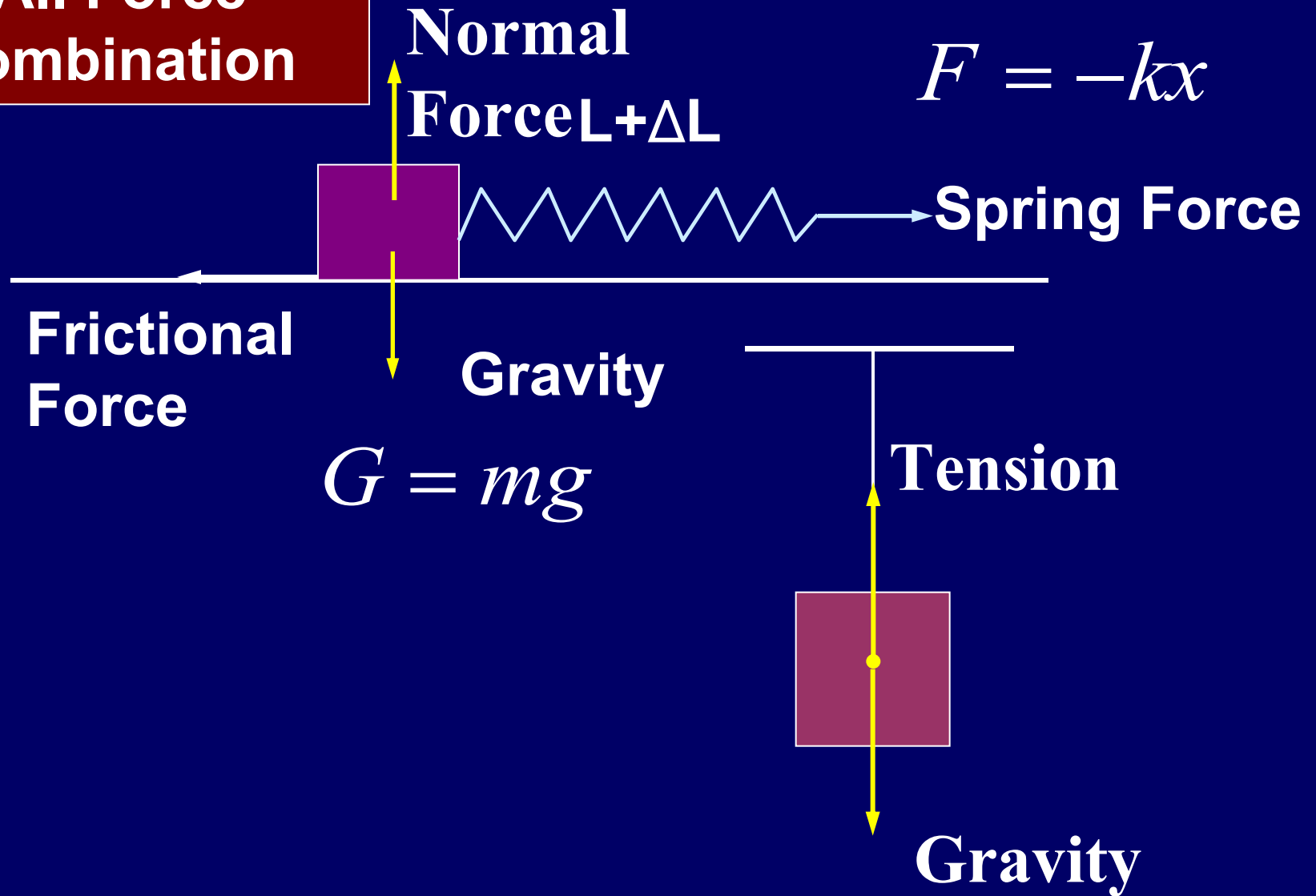
**With ABS, the car stop
in shorter distance**



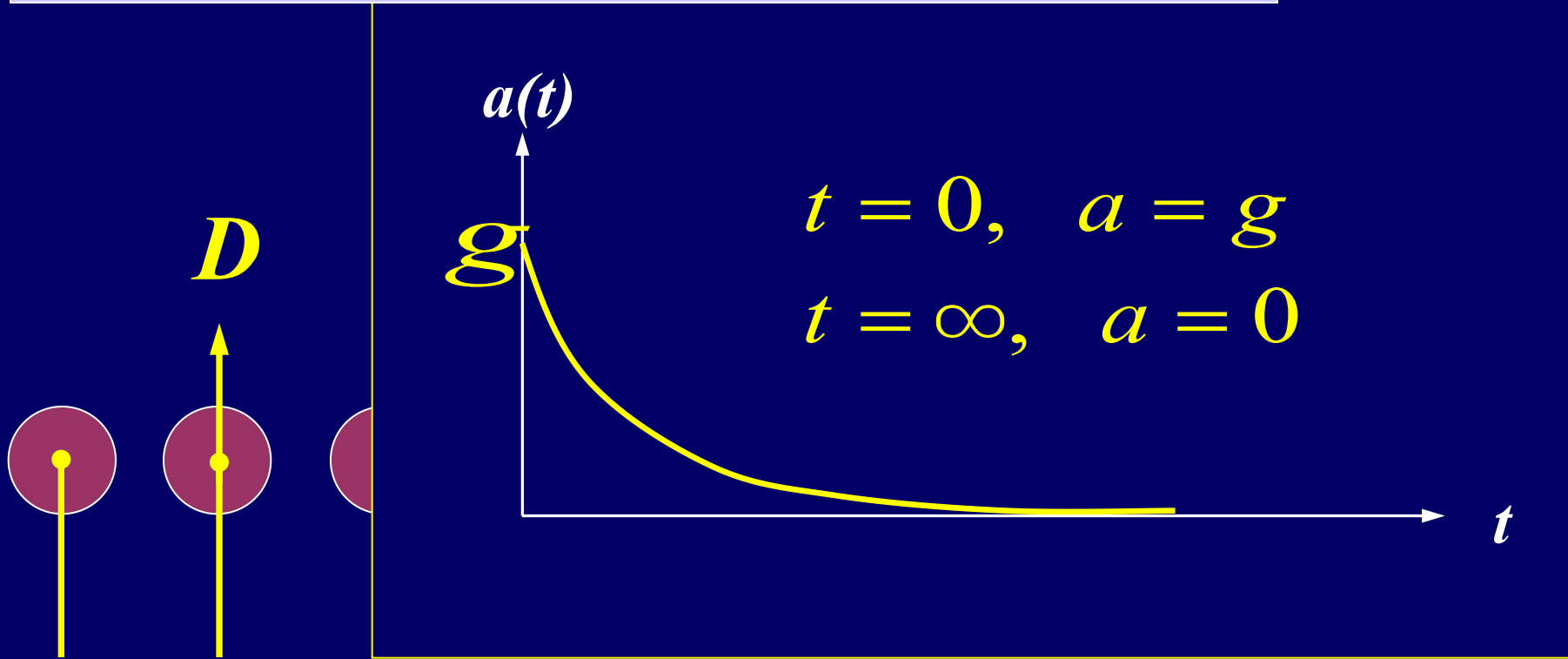
The direction of frictional force?



**All Force
combination**



force depending on the velocity.



$$a(t) = \frac{dv(t)}{dt} \quad 1 - (b / mg)v = e^{-bt/m}$$

$$a(t) = ge^{-bt/m} \quad v(t) = \frac{mg}{b} (1 - e^{-bt/m})$$

force depending on the velocity.

Position dependent forces

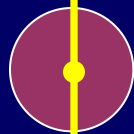
$$F = ma$$

Time dependent forces

$$m\sigma - bv = m \frac{dv}{dy} = m \frac{dv}{dy} \frac{dy}{dt} = mv \frac{dv}{dy}$$

$$dy = \frac{mv dv}{mg - bv} = \left[\frac{(1 - bv/mg)b/m}{(1 - bv/mg)b/m} - \frac{1}{b/m} \right] dv$$

$$y = -\frac{m^2 g}{b^2} \ln\left[1 - \frac{bv}{mg}\right] - \frac{mv}{b}$$



mg

mg

mg

Force dependent on the position

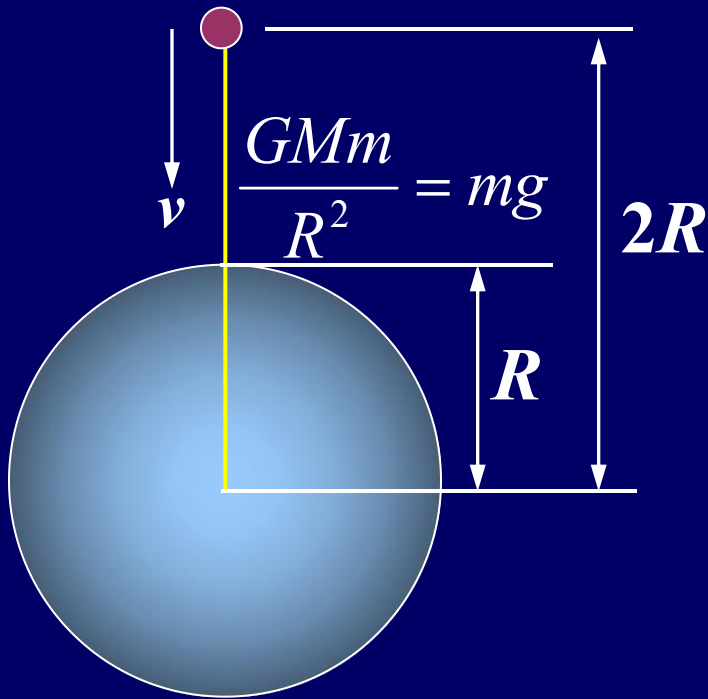
$F=F(r)$. Thus a depends on the position, i.e, $a(r)$.

For one dimension, $a=a(x)$

$$a(x) = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx} \quad \therefore a(x)dx = vdv;$$

$$\int_{x_0}^x a(x)dx = \int_{v_0}^v vdv = \frac{1}{2}(v^2 - v_0^2)$$

Example An object, located at $2R$ away from the center of the earth, is falling from rest vertically. R is the radius of the earth. Find the speed when it arrives at the earth's surface. The drag force on the object by the air and the spin of the earth are ignorable.



$$F = G \frac{Mm}{r^2} = ma$$

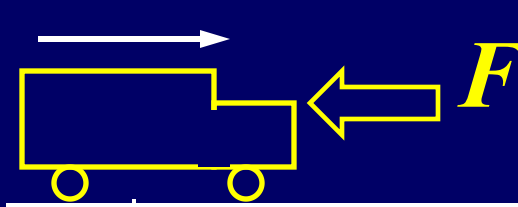
$$G \frac{M}{r^2} = -v \frac{dv}{dr}$$

$$\int_{2R}^R -G \frac{M}{r^2} dr = \int_0^v v dv$$

$$gR = G \frac{M}{R} = v^2 \quad v = \sqrt{gR}$$

Time dependent forces

Example: A car moves with the velocity v_0 along the x axis. If $a(t) = -ct$, how much time it passes and how far it travels when it stops?



The diagram shows a car represented by a rectangle on wheels, moving to the right as indicated by a long arrow above it. A force F is shown as a shorter arrow pointing to the left, acting on the car. Below the car, a horizontal line represents the ground, with a point labeled x_0 marked by a vertical tick.

$$a(t) = -ct, \quad a = \frac{dv}{dt} \quad \xrightarrow{x} \quad dv = a dt$$

$$x(t) = x_0 + \int_0^t v(t) dt$$

$$v(t) = v_0 - \int_0^t c t dt$$

$$x(t) = x_0 + v_0 t - \frac{1}{6} c t^3$$

$$v(t) = v_0 - \frac{1}{2} c t^2$$

$$x(t) = x_0 + \frac{2v_0}{3} \sqrt{\frac{2v_0}{c}} \quad t = \sqrt{\frac{2v_0}{c}} \quad 0 = v_0 - \frac{1}{2} c t^2$$

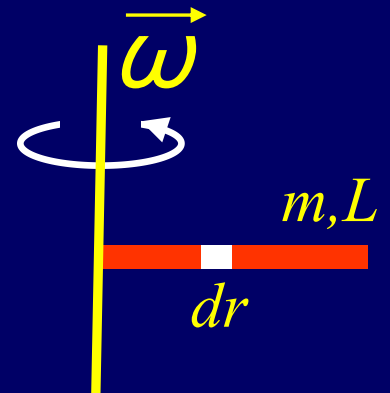
Example: A uniform rod with the mass m and the length L is perpendicularly fixed on the rotating axis. When it rotates with a angular speed ω , find the tension force of the rod at every point if the gravitation force is ignorable.

Answer:
$$F(r) - F(r + dr) = r\omega^2 dm$$

$$dF(r) = F(r + dr) - F(r) = -r\omega^2 dm$$

$$\int_r^L dF(r) = -\int_r^L \frac{m\omega^2}{L} r dr$$

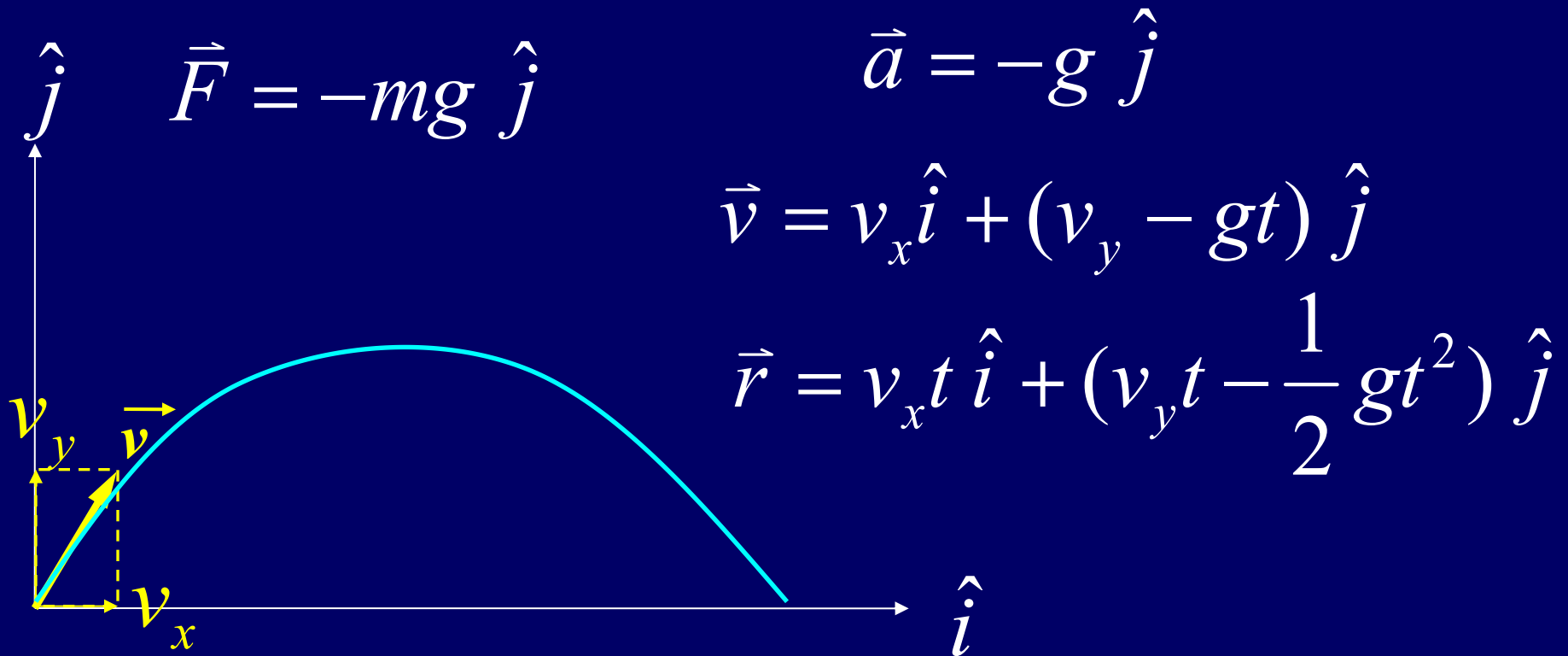
$$F(L) = 0 \qquad F(r) = \frac{m\omega^2}{2L} (L^2 - r^2)$$



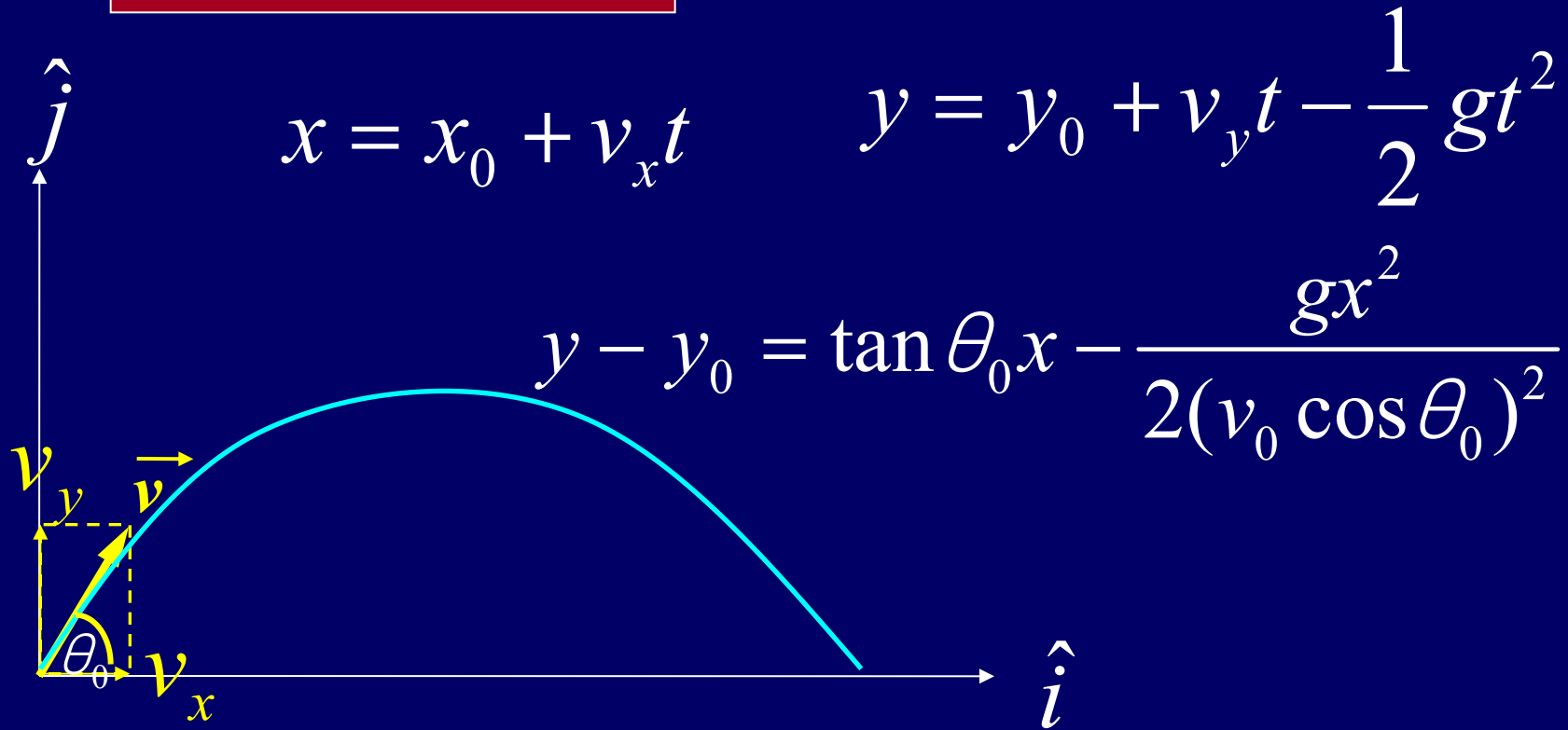
Chap. 4

MOTION IN 2- and 3-DIMENSIONS

Projectile motion-two dimension



Path Equation:

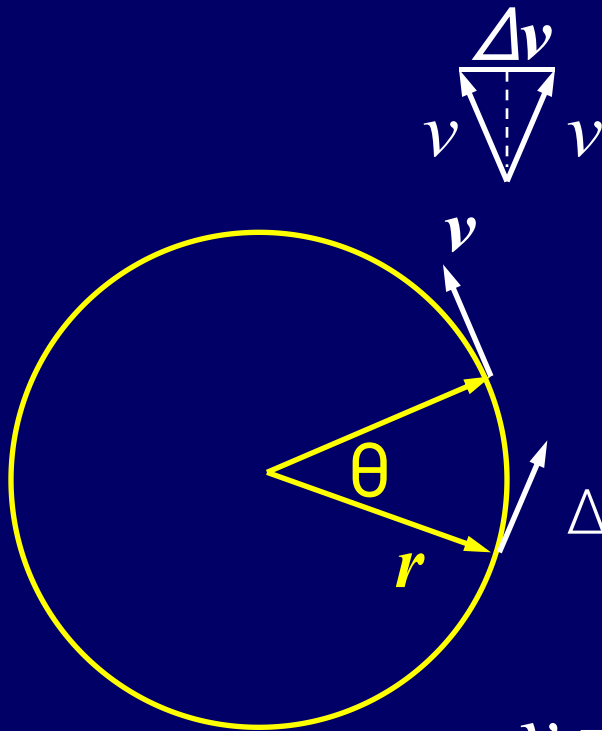


Horizontal Range:

$$y - y_0 = 0$$

$$R = (v_0^2 / g) \sin 2\theta_0$$

Uniform Circular motion



$$r\Delta\theta = v\Delta t \quad \Delta t = \frac{r\Delta\theta}{v}$$

$$2v \sin \frac{\Delta\theta}{2} = \Delta v$$

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{2v \sin \frac{\Delta\theta}{2}}{\frac{r\Delta\theta}{v}} = \frac{v^2}{r} \frac{\sin \frac{\Delta\theta}{2}}{\frac{\Delta\theta}{2}}$$

$$\Delta t \rightarrow 0, \Delta\theta \rightarrow 0, \frac{\sin \frac{\Delta\theta}{2}}{\frac{\Delta\theta}{2}} = 1 \quad a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{v^2}{r}$$

Tangential Acceleration

$$v \neq \text{const.} \quad \frac{dv}{dt} \neq 0 \quad \vec{a} = \vec{a}_c + \vec{a}_t$$

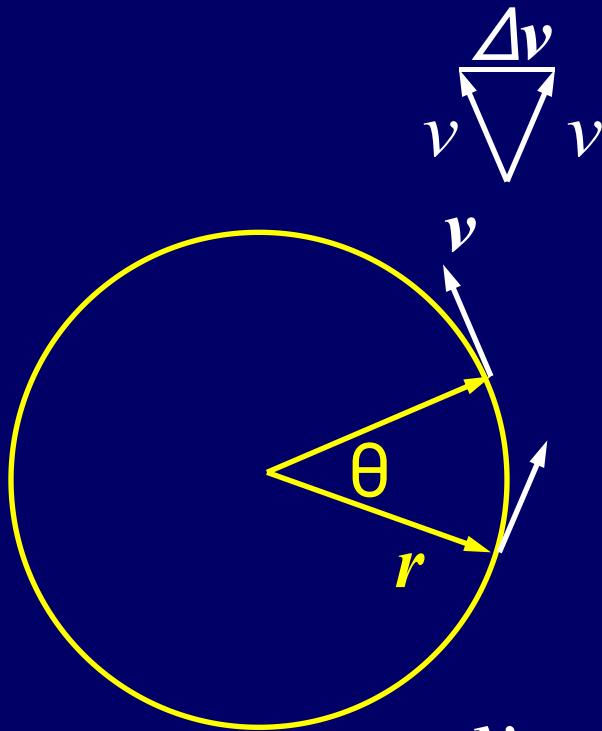
$$a_c = \frac{v^2}{r}$$

$$a_t = \frac{dv}{dt}$$

Centripetal Acceleration

Uniform Circular motion

$$\vec{v} = v\hat{u}_\phi$$



$$\begin{aligned}\vec{a} &= \frac{d\vec{v}}{dt} = \frac{d}{dt}(v\hat{u}_\phi) = v\frac{d\hat{u}_\phi}{dt} + \frac{dv}{dt}\hat{u}_\phi \\ &= -v\frac{d\phi}{dt}\hat{u}_r + \frac{dv}{dt}\hat{u}_\phi = -v\frac{d(\frac{s}{r})}{dt}\hat{u}_r + \frac{dv}{dt}\hat{u}_\phi \\ &= -\frac{v^2}{r}\hat{u}_r + \frac{dv}{dt}\hat{u}_\phi\end{aligned}$$

Tangential Acceleration

$$v = \text{const.} \quad \frac{dv}{dt} = 0 \quad \vec{a} = \vec{a}_c + \vec{a}_T$$

$$a_c = \frac{v^2}{r}$$

$$a_T = \frac{dv}{dt}$$

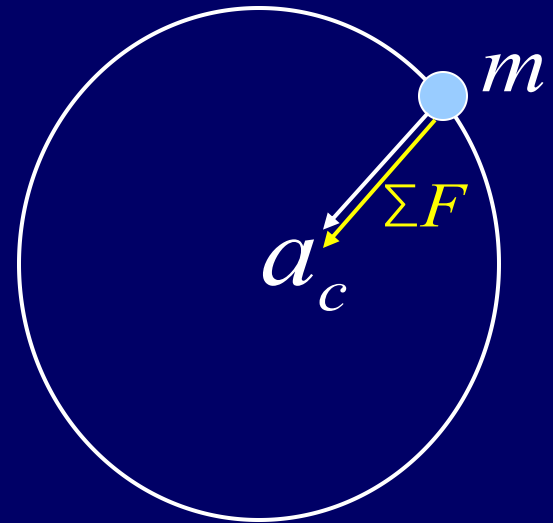
Centripetal Acceleration

The Dynamics of Uniform Circular Motion

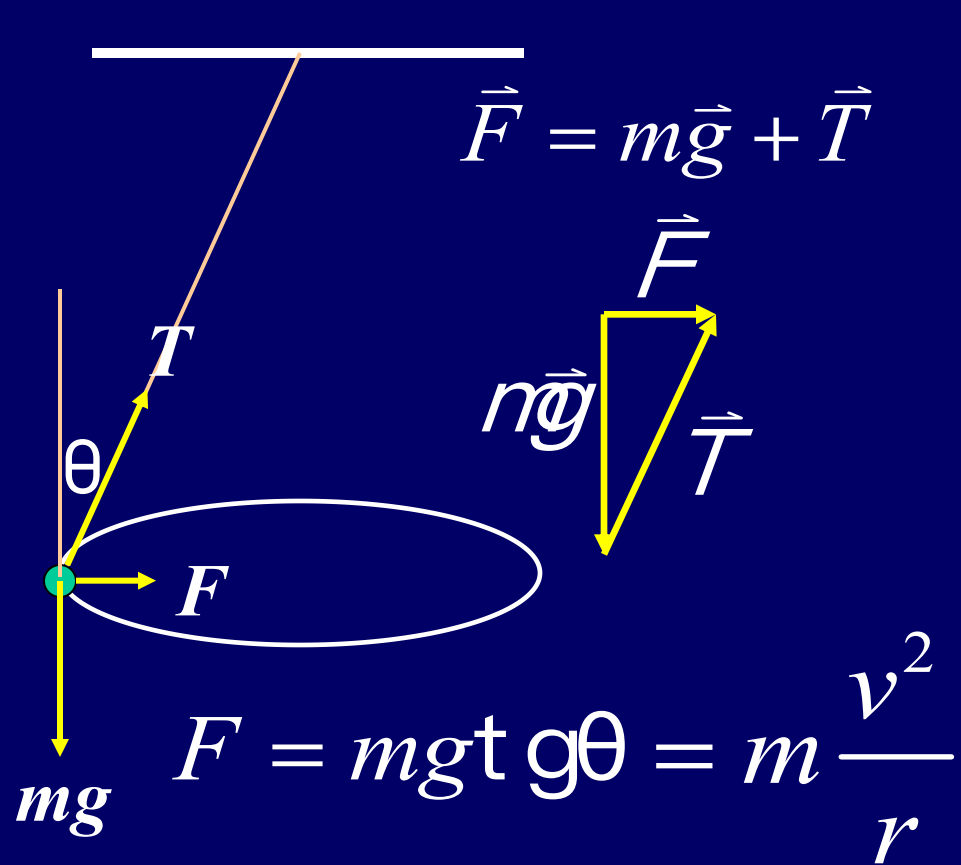
Centripetal acceleration

$$a_c = \frac{v^2}{r}$$

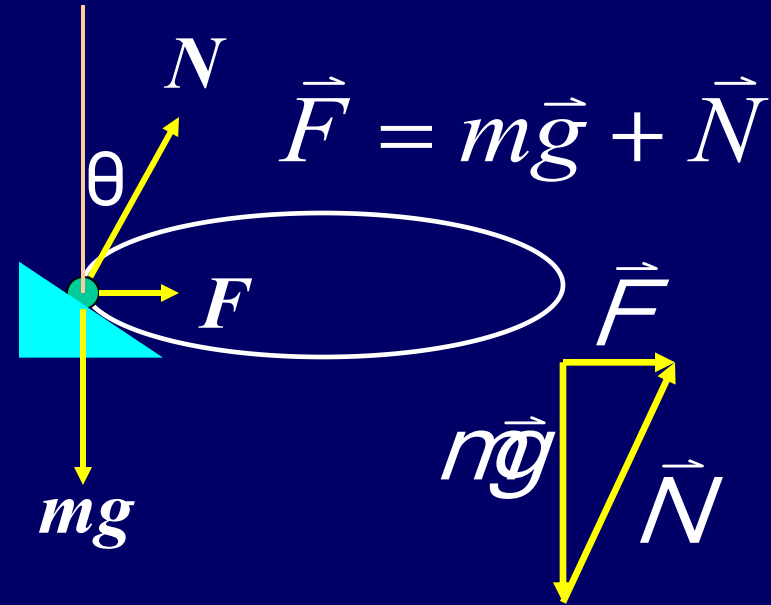
$$\Sigma F = m \frac{v^2}{r}$$



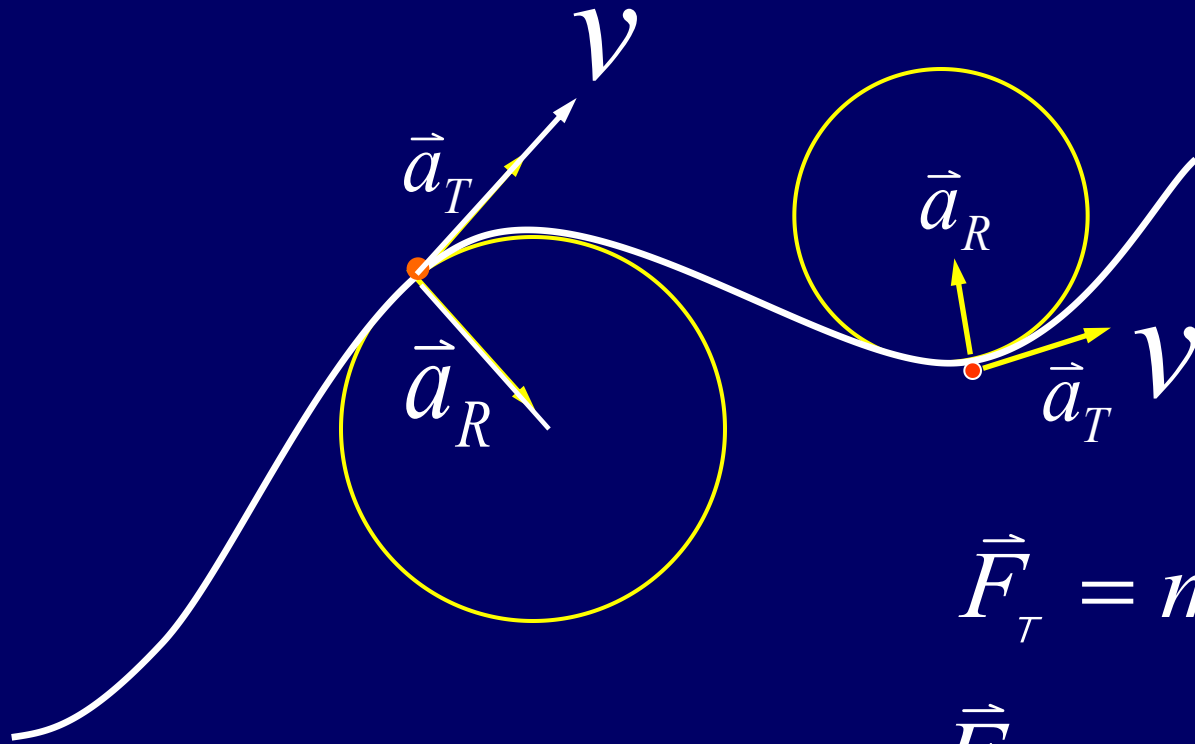
pendulum



Banked curve



In Nature Coordinates



$$\vec{F}_T = m\vec{a}_T$$

$$\vec{F}_R = m\vec{a}_R$$

Example The position vector of a particle is $\vec{r} = (2t^2 - 5)\hat{i} + (t^2 + 3t)\hat{j}$

What is the tangential and centripetal acceleration, at $t=0$?

$$\vec{r} = (2t^2 - 5)\hat{i} + (t^2 + 3t)\hat{j}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = 4t\hat{i} + (2t + 3)\hat{j}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = 4\hat{i} + 2\hat{j}$$

$$a_T = \frac{dv}{dt} = \frac{20t + 6}{\sqrt{(4t)^2 + (2t + 3)^2}}$$

$$a_R = \frac{v^2}{\rho} = \omega^2 \rho$$

$$v = \sqrt{(4t)^2 + (2t + 3)^2}$$

$$a_T^2 + a_R^2 = a^2 \quad a_R = \sqrt{a^2 - a_T^2}$$

$$a_R|_{t=0} = \sqrt{20 - 4} = 4$$

$$a_T|_{t=0} = 2$$

How to calculate the radial and poloidal velocity?

Another simple way to calculate the tangential and centripetal acceleration

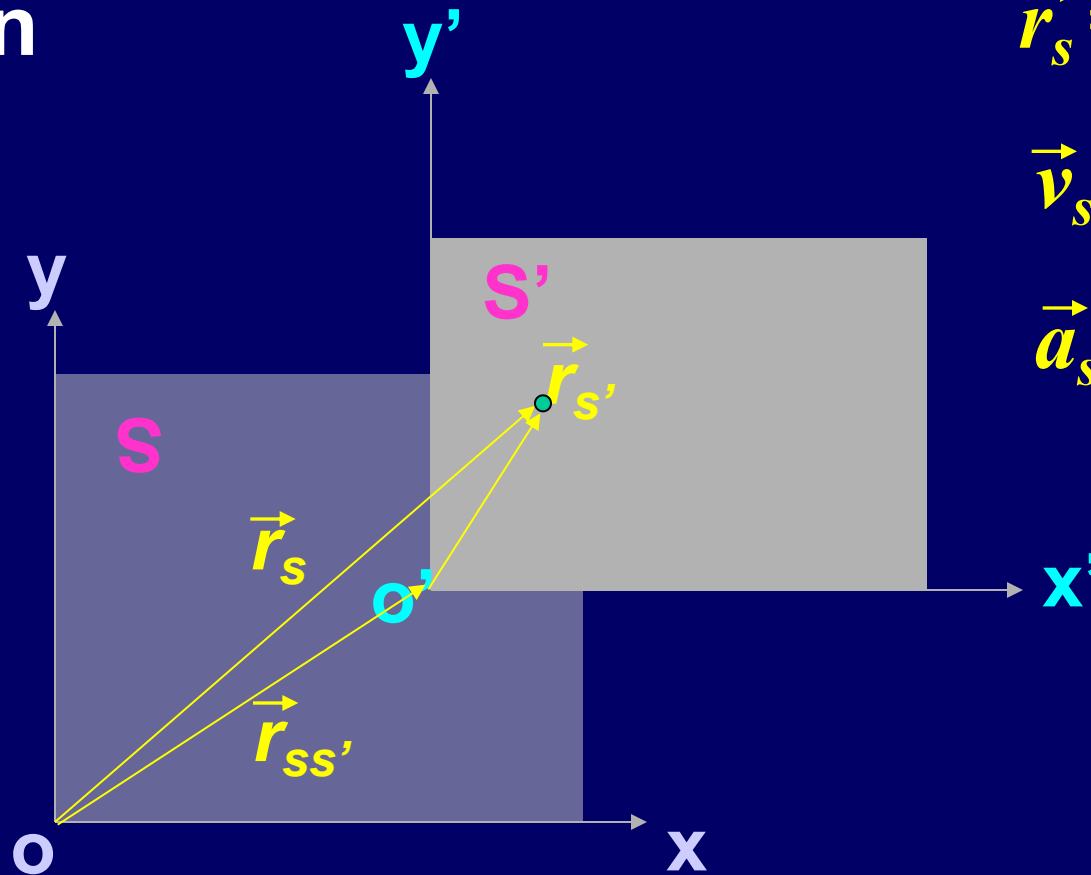
$$\vec{v}_r = (\vec{v} \cdot \hat{r})\hat{r} \quad \vec{v}_\theta = \vec{v} \cdot (\vec{I} - \hat{r}\hat{r})$$

$\hat{r} = \vec{r} / r$ is a unit vector

$$\vec{a}_t = (\vec{a} \cdot \hat{v})\hat{v} \quad \vec{a}_R = \vec{a} - \vec{a}_t$$

$\hat{v} = \vec{v} / v$ is a unit vector

Relative motion



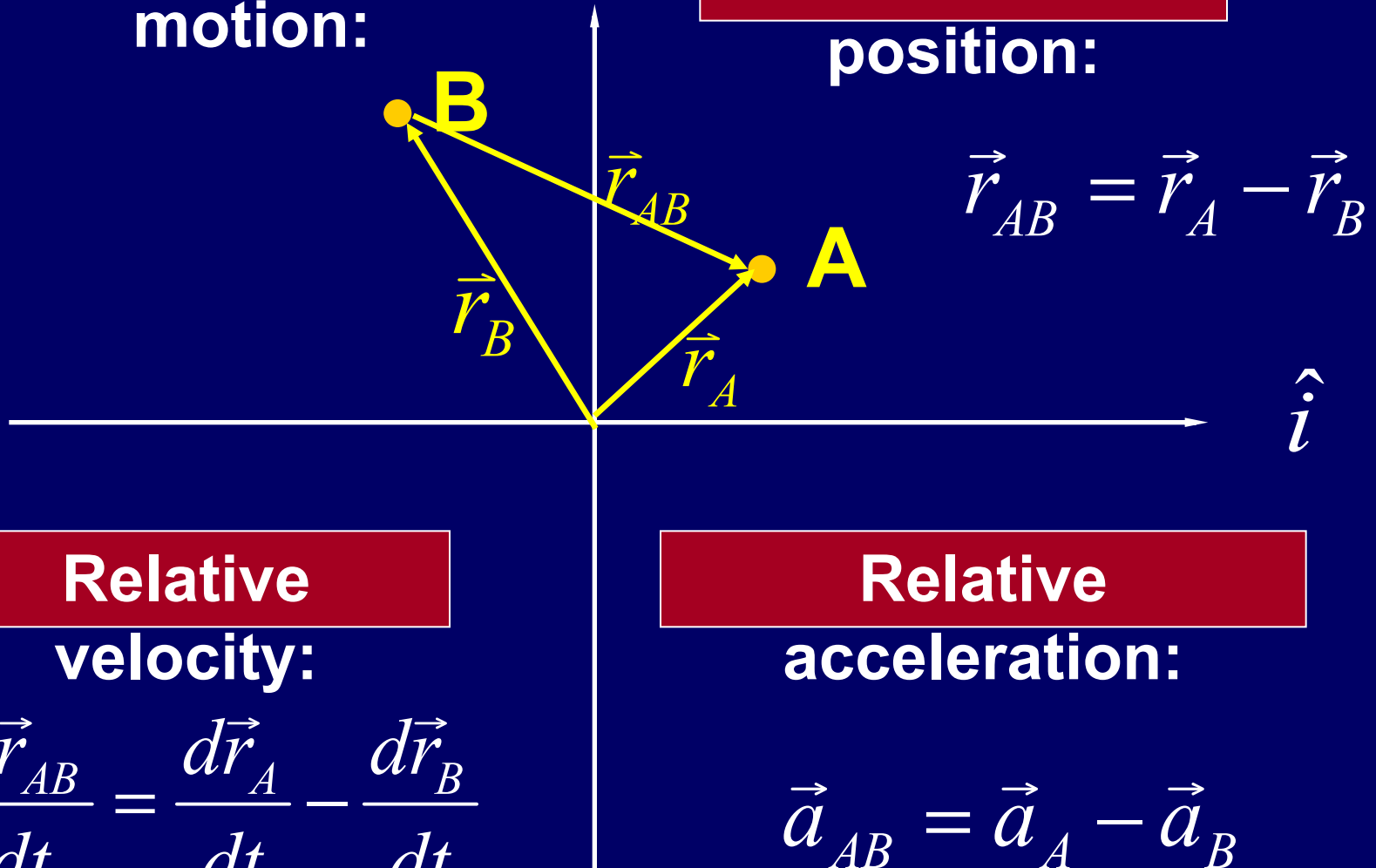
$$\vec{r}_s = \vec{r}_{ss'} + \vec{r}_{s'}$$

$$\vec{v}_s = \vec{v}_{ss'} + \vec{v}_{s'}$$

$$\vec{a}_s = \vec{a}_{ss'} + \vec{a}_{s'}$$

**Relative
motion:**

**Relative
position:**



**Relative
velocity:**

**Relative
acceleration:**

$$\frac{d\vec{r}_{AB}}{dt} = \frac{d\vec{r}_A}{dt} - \frac{d\vec{r}_B}{dt}$$

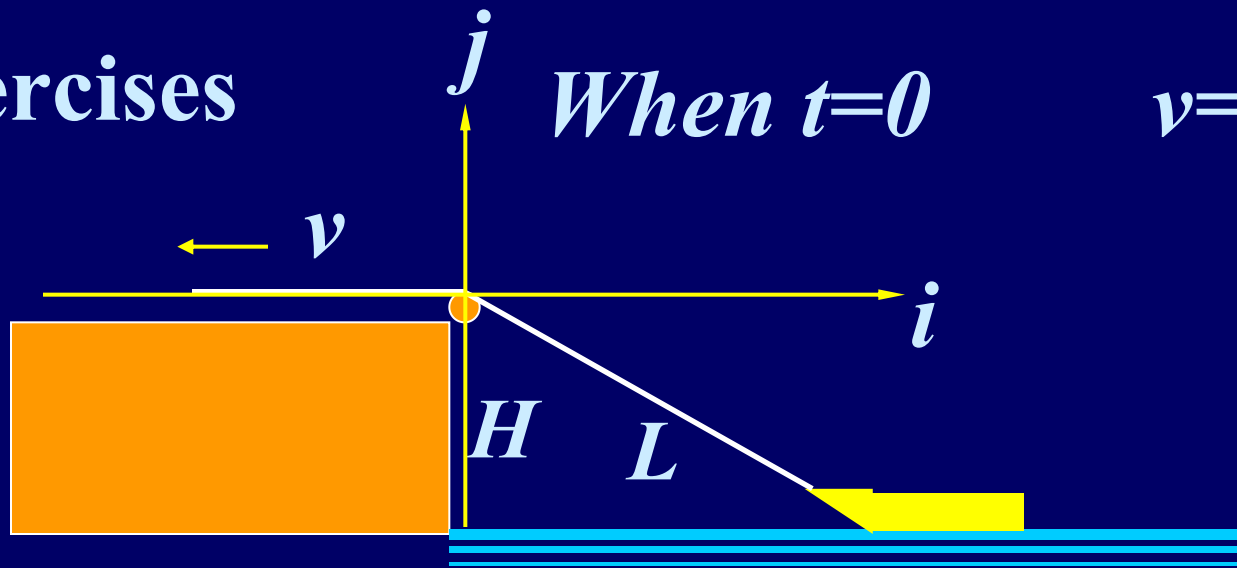
$$\vec{a}_{AB} = \vec{a}_A - \vec{a}_B$$

$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$$

Exercises

When $t=0$

$v=?$ $a=?$



$$\vec{r} = \sqrt{(L-vt)^2 - H^2} \vec{i} - H \vec{j}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{-(L-vt)v}{\sqrt{(L-vt)^2 - H^2}} \vec{i} \quad \vec{v}|_{t=0} = \frac{d\vec{r}}{dt} = \frac{-Lv}{\sqrt{L^2 - H^2}} \vec{i}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{-H^2 v^2}{[(L-vt)^2 - H^2]^{3/2}} \vec{i} \quad \vec{a}|_{t=0} = \frac{d\vec{v}}{dt} = \frac{-H^2 v^2}{(L^2 - H^2)^{3/2}} \vec{i}$$

Example: The flow speed in a river with the width l is linearly dependent on the distance away from the bank of the river. The maximum flow speed is v_0 . A boat with the speed v departs from a bank. When the boat arrives at the position that is $l/4$ away from the bank, there is no enough fuel to reach another bank for the boat. Then, the boat return back the original bank with the speed $v/2$ and the direction perpendicular to the flow velocity. Find how far away from the departed position when the boat arrives the bank.

Answer:

The boat velocity is $\frac{dx}{dt} = \frac{2v_0}{l} y \quad \frac{dy}{dt} = v$

We have $dx = \frac{2v_0}{lv} y dy$, then $\int_0^x dx = \int_0^y \frac{2v_0}{lv} y dy$

The boat trajectory is $x = \frac{v_0}{lv} y^2$, thus $y_0 = \frac{l}{4}$, $x_0 = \frac{v_0 l}{16v}$

The returning boat velocity is $\frac{dx}{dt} = \frac{2v_0}{l} y$ $\frac{dy}{dt} = -v/2$

We have $dx = -\frac{4v_0}{lv} y dy$, then $\int_{x_0}^x dx = \int_{y_0}^0 -\frac{4v_0}{lv} y dy$

The boat position: $x = \frac{3lv_0}{16v}$

Exercises

P63 7, 9

P86 15, 18, 20

P87 23